Novel catalysts for selective oxidation of *n*-butane: Structure-performance studies and characterization for a newly designed lab-scale profile reactor

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M.Sc.
Rhea Machado

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Promotionsausschuss:

Vorsitzender: Prof. Dr. Jens-Uwe Repke
Gutachter: Prof. Dr. Matthias Kraume
Gutachter: Prof. Dr. Reinhard Schomäcker
Gutachter: Prof. Dr. Raimund Horn

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To my beloved grandfather, Chrestos, whose wisdom, guidance, and enduring love have shaped my perspective on the world. This is for you.

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Abstract

The direct conversion of light alkanes to oxygenates in heterogeneously catalyzed gas-phase reactions remains a challenge primarily driven by identifying novel solids with appropriate catalytic functionalities. The likelihood of success can be increased by contriving profound synthesis strategies. Additionally, such catalysts' property performance correlations can help enhance the knowledge of selective oxidation reactions. Using various in-situ characterization techniques and combining their results with catalytic investigations, new descriptors for oxidation to oxygenates can be discovered.

The first focus of this work is the characterization for a newly designed lab-scale profile reactor for spatially resolved high-resolution profiles of temperature and species and a preliminary structure and composition study of two innovative catalyst classes for the selective oxidation of *n*-butane to maleic anhydride (MAN).

Tracking intermediates along the catalytic bed of new catalyst classes would help study their chemical reaction networks in detail. Traditional tube setups accomplish this by measuring different catalyst masses at constant gas flow or increasing gas velocity. Drawbacks of these measurements are the high time and catalyst material consumption (8 points – 8 reactors) or changed reaction conditions.[1] It is possible to study kinetics in an alternative way with profile reactors. Many kinetic data are available with one profile without changing temperature or gas velocity. Those relatively big data sets can be used to decode reaction networks and verify or feed kinetic simulations. This information can provide significant input for the design and operation of industrial reactors.

The size of the profile reactor was chosen to fit typical amounts from lab-scale catalyst synthesis, usually < 1g hence an ideal tool for investigating novel materials. One of the main challenges was designing a setup that prevents the polymerization of MAN at any location. By applying a kinetic model from the literature [2], the species profiles resulting from partial oxidation of *n*-butane over VPP catalyst could be predicted for reactants and main products (CO, CO_2, and MAN); consequently, the profile reactor behaved as expected and demonstrated the proof of concept. Focusing on the minor products acetic acid and acrylic acid, the model could correctly predict the concentration ranges but less accurately their

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reaction progress along the catalytic bed. Consequently, new parameter estimation was considered. Therefore, a novel open-source kinetic modeling tool written in Python was developed. This tool is based on the global search algorithm Differential Evolution and can optimize the kinetic parameters without an initial guess. The newly estimated parameters showed an overall better fit regarding the species profiles and are also in accordance with the literature [3–5].

In cooperation with the group of Prof. Glaum (University of Bonn) and the group of Stephan A. Schunk (hte GmbH), two new selective oxidation catalyst classes were synthesized by employing two different strategies. Additionally, these catalyst classes were synthesized on the one hand via solution combustion synthesis by the group of Prof. Glaum to obtain unsupported catalysts with a certain crystallinity and, on the other hand, via incipient wetness impregnation by hte GmbH to obtain supported catalysts with a potentially more comprehensive compositional range. The focus of this work was to evaluate these new catalyst classes and to study the interrelations between crystallinity, elemental composition, and catalytic performance.

The first synthesis strategy appropriated the knowledge of key physiochemical properties of vanadyl pyrophosphate (VPP) and new insights of recently published catalyst class $V_{1-x}W_xOPO_4$ with an α_{II}-$VOPO_4$-related phase.[6] The idea of the new approach was to incorporate the well-known VPP promoter Nb in the α_{II}-$VOPO_4$-related phase, which resulted in unsupported $V_xNb_{1-x}OPO_4$. The compositional study showed that $V_{0.1}Nb_{0.9}OPO_4$ showed the best performance with a MAN yield of about 23% (X = 84%). The nearly XRD amorphous supported $xV(1-x)NbPO_y/SiO_2$ exhibited until a V fraction of x = 0.7 only diffraction reflexes of α_I-$VOPO_4$-related phase, hence differed significantly from the unsupported catalysts. Lower MAN yields were achieved with the supported catalysts, and an opposite behavior regarding elemental composition was observed; high V fractions led to better performance.

The second strategy used platinum group metal (PGMs) catalysts' known ability to activate the C-H bonds of alkanes. Catalysts based on the incorporation of metallic PGMs have presented themselves as efficient total combustion catalysts. The transformation to a selective oxidation catalyst might be achieved by diluting PGM cations in a well-defined oxide host structure. This approach led to the synthesis of isolated Ru-based active sites in a

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tungsten phosphate matrix appearing as a ReO_3-type structure $(Ru_aW_bP_c)O_{2.5+\delta}$. However, the limitation is that a single-phase ReO_3-type structure can only be achieved for a small compositional range. XRD amorphous ruthenium tungsten phosphates supported on SiO_2 were synthesized to study the whole compositional range. Unsupported crystalline and supported XRD amorphous phases with similar Ru/W/P compositions showed comparable performance. XRD and TEM investigations confirmed a ReO_3-related structure for the unsupported $(Ru_{0.1}W_{0.4}P_{0.5})O_{2.5+\delta}$ catalysts, and there are at least strong indications that supported ruthenium tungsten phosphates may adopt a ReO_3-related structure as well. Elemental composition studies with supported catalysts showed that the tuning of Ru and W content led to higher MAN yields; hence composition seems to be more crucial than the degree of crystallinity. This new class can be presented as the first selective oxidation catalyst to oxygenates without incorporating V and Mo.

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Zusammenfassung

Die direkte Umwandlung von kurzkettigen Alkanen in Oxygenate in heterogen katalysierten Gasphasenreaktionen bleibt eine Herausforderung. Diese wird hauptsächlich durch die Identifizierung neuer Feststoffe mit geeigneten katalytischen Funktionalitäten vorangetrieben. Die Erfolgswahrscheinlichkeit kann durch die Entwicklung fundierter Synthesestrategien erhöht werden. Zusätzlich können Eigenschafts-Leistungs-Korrelationen solcher Katalysatoren dazu beitragen, das Wissen über selektive Oxidationsreaktionen zu erweitern. Durch die Verwendung verschiedener In-situ-Charakterisierungstechniken und die Kombination ihrer Erkenntnisse mit denen aus katalytischen Untersuchungen könnten neue Deskriptoren für die selektive Oxidation zu Oxygenaten erschlossen werden.

Der Schwerpunkt dieser Arbeit liegt auf die Charakterisierung eines neu entwickelten Profilreaktors im Labormaßstab für hochortsaufgelöste Temperatur- und Speziesprofile. Und einer vorläufigen Struktur- und Zusammensetzungsstudie zweier innovativer Katalysatorklassen für die selektive Oxidation von *n*-Butan zu Maleinsäureanhydrid (MAN).

Die Verfolgung von Zwischenprodukten entlang des Katalysatorbetts dieser neuen Katalysatorklassen würde helfen, ihre chemischen Reaktionsnetzwerke im Detail zu untersuchen. Herkömmliche Testreaktoren erreichen dies, indem sie unterschiedliche Katalysatormassen bei konstantem Gasfluss oder steigender Gasgeschwindigkeit bei konstanter Masse messen. Nachteile dieser Messungen sind der hohe Zeit- und Katalysatormaterialverbrauch (8 Punkte – 8 Reaktoren) oder veränderte Reaktionsbedingungen.[1] Es ist möglich, die Kinetik auf alternative Weise mit Profilreaktoren zu untersuchen. Viele kinetische Daten sind mit einem Profil verfügbar, ohne Temperatur oder Gasgeschwindigkeit zu ändern. Diese relativ großen Datensätze können verwendet werden, um Reaktionsnetzwerke zu entschlüsseln und kinetische Simulationen zu verifizieren oder zu füttern.

Die Größe des Profilreaktors wurde so gewählt, dass typische Mengen aus der Katalysatorsynthese im Labormaßstab passen, normalerweise < 1 g, daher eine ideale Methode für die Untersuchung neuer Materialien. Eine der größten Herausforderungen

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bestand darin, einen Aufbau zu entwerfen, der die Polymerisation von MAN an jedem Ort verhindert. Durch Anwendung eines kinetischen Modells aus der Literatur [2] konnten die aus der partiellen Oxidation von n-Butan über Vanadylpyrophosphat-Katalysator resultierenden Speziesprofile für Reaktanten und Hauptprodukte (CO, CO_2 und MAN) vorhergesagt werden; folglich verhielt sich der Profilreaktor wie erwartet und demonstrierte den Konzeptnachweis. Die Konzentrationsbereiche der Nebenprodukte Essigsäure und Acrylsäure konnten durch das Model korrekt vorhergesagt werden, aber den Reaktionsfortschritt entlang des Katalysatorbetts weniger genau. Folglich wurde eine neue Parameterschätzung in Betracht gezogen. Daher wurde ein neues Open-Source-Tool zur Modellierung von Kinetik entwickelt, das in Python geschrieben ist. Dieses Tool basiert auf dem globalen Suchalgorithmus Differential Evolution und kann die kinetischen Parameter ohne anfängliche Vermutung optimieren. Die neu geschätzten Parameter zeigten eine insgesamt bessere Passung bezüglich der Artenprofile und stimmen auch mit der Literatur überein [3–5].

In Zusammenarbeit mit der Gruppe von Prof. Glaum (Universität Bonn) und der Gruppe von Stephan A. Schunk (hte GmbH) wurden zwei neue Katalysatorklassen für die selektive Oxidation synthetisiert. Zusätzlich wurden diese Katalysatorklassen einerseits durch Lösungsverbrennungssynthese durch die Gruppe von Prof. Glaum synthetisiert, um Vollkatalysatoren mit einer bestimmten Kristallinität zu erhalten, und andererseits durch Anfangs- Nassimprägnierung durch die hte GmbH, um Trägerkatalysatoren mit einem möglicherweise umfassenderen Zusammensetzungsbereich zu erhalten. Der Schwerpunkt dieser Arbeit lag auf der Bewertung dieser neuen Katalysatorklassen und der Untersuchung der Wechselbeziehungen zwischen Kristallinität, elementarer Zusammensetzung und katalytischer Performance.

Die erste Synthesestrategie nutzte das Wissen um die wichtigsten physikalisch-chemischen Eigenschaften von Vanadylpyrophosphat (VPP) und neue Erkenntnisse der kürzlich veröffentlichten Katalysatorklasse $V_{1-x}W_xOPO_4$ mit einer α_{II}-$VOPO_4$- verwandten Phase.[6] Die Idee des neuen Ansatzes bestand darin, den wohlbekannten VPP-Promotor Nb in die α_{II}-$VOPO_4$- verwandten Phase einzubauen, was zu Vollkatalysatoren mit einer α_{II}-$VOPO_4$-ähnlichen Phase führte. Die Zusammensetzungsstudie zeigte, dass $V_{0.1}Nb_{0.9}OPO_4$ die beste Leistung mit einer MAN-Ausbeute von etwa 23 % (X = 84 %) aufwies. Die nahezu XRD-amorphen geträgerten $xV(1-x)NbPO_y/SiO_2$ zeigten bis zu einem V-Anteil von x = 0,7 nur

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Beugungsreflexe der α_I-$VOPO_4$-verwandten Phase und unterschieden sich daher signifikant von den Vollkatalysatoren. Mit den geträgerten Katalysatoren wurden niedrigere MAN-Ausbeuten erzielt, und es wurde ein entgegengesetztes Verhalten bezüglich der Elementzusammensetzung beobachtet; hohe V-Anteile führten zu einer besseren Leistung.

Die zweite Strategie nutzte die bekannte Fähigkeit von Platingruppenmetall(PGMs)-Katalysatoren, die C-H-Bindungen von Alkanen zu aktivieren. Katalysatoren, die auf dem Einbau von metallischen PGMs basieren, haben sich als wirksame Katalysatoren für die vollständige Verbrennung von Kohlenwasserstoffen erwiesen. Die Umwandlung in einen selektiven Oxidationskatalysator könnte durch Verdünnen von PGM-Kationen in einer wohldefinierten Oxid-Wirtsstruktur erreicht werden. Dieser Ansatz führte zur Synthese isolierter Ru-basierter aktiver Zentren in einer Wolframphosphatmatrix die als Struktur vom ReO_3-Typ $(Ru_aW_bP_c)O_{2.5+\delta}$ aufweisen. Die Einschränkung besteht jedoch darin, dass eine einphasige Struktur vom ReO_3-Typ nur für einen kleinen Zusammensetzungsbereich erreicht werden kann. XRD-amorphe Ruthenium-Wolfram-Phosphate auf SiO_2-Trägern wurden synthetisiert, um den gesamten Zusammensetzungsbereich zu untersuchen. Kristalline Vollkatalysatoren und geträgerte amorphe XRD-Phasen mit ähnlichen Ru/W/P-Zusammensetzungen zeigten eine vergleichbare Performance. XRD- und TEM-Untersuchungen bestätigten eine ReO_3-verwandte Phase für die Vollkatalysatoren $(Ru_{0.1}W_{0.4}P_{0.5})O_{2.5+\delta}$, und es gibt zumindest starke Hinweise darauf, dass geträgerte Ruthenium-Wolfram-Phosphate auch eine ReO_3-verwandte Phase aufweisen. Elementzusammensetzungsstudien mit geträgerten Katalysatoren zeigten, dass die Abstimmung des Ru- und W-Gehalts zu höheren MAN-Ausbeuten führte; daher scheint die Zusammensetzung entscheidender zu sein als der Kristallinitätsgrad. Diese neue Klasse kann als erster selektiver Oxidationskatalysator für Oxygenate ohne Einbau von V und Mo präsentiert werden.

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1 Introduction and objectives

The utilization of light alkanes as a feedstock material for the chemical industry is a promising option from both economic and environmental sustainability points of view [7]. Small alkanes can be derived from natural gas composed mainly of methane and other lower alkanes or as side products that are part of the streams within petrochemical or bio-refineries; alkanes of interest are namely ethane, propane, and butane. In recent years, selective C-H activation of these molecules has received intense attention from academia and industry.[8, 9]

The selective oxidation of *n*-butane to maleic anhydride (MAN) is a crucial industrial process within the value-added chain of chemical manufacturing. The global production of MAN has reached a volume of around 3 million metric tons per year.[10]

Since the first plant start-up in the 1960s, the molar yield has been constantly improved. However, the state-of-the-art MAN production plants are running with yields below 65 %, and no significant breakthrough has been achieved recently.[11–13]

Industrially, vanadyl pyrophosphate (VPP) is used as a catalyst to carry out the oxyfunctionalization of *n*-butane to MAN. Although significant effort has been undertaken by modifying VPP or exploring reaction conditions, the scientific community agrees that no substantial enhancement will be realized with this catalyst in the near future.[6]

This resulted in realigning of research focus from improving the VPP catalyst. The focus should be on developing new catalyst classes to break the yield limitation; therefore, a holistic development strategy is essential. New synthesis concepts should be framed by considering heterogeneous and homogeneous catalysis descriptors.[14]

Without a deep understanding of the catalytic process, no effective development of new or existing catalysts will occur. Especially the knowledge of catalyst active sites and their interactions with reactants, intermediates, and products is fundamental.

Next to using a wide range of characterization techniques, a smart reactor design should provide detailed insights into the catalytic behavior [15, 16]. The concept of profile reactors was coined and pushed in the last two decades. This relatively new concept allows

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simultaneously analyzing the gas composition, temperature, and catalyst state at each point within the catalytic bed.[17, 18]

Hence spatially resolved high-resolution species profiles are created, and the black box behavior of conventional fixed bed reactor setup is overcome. A profile reactor not only supports the development of new catalysts but can also be used as a tool for industrial reactor design. Of significant importance for the chemical industry are lean downstream processes. The fewer unwanted side products, the cheaper the desirable products' purification is. With profile reactors, it is possible to understand when and how the side products are formed within the bed. With this knowledge, new strategies can be developed to avoid the formation of side products or to combust these without influencing the selectivity of the desired products. One possible strategy is the design of multi-functional catalyst beds by combining at least two different catalysts. Profile reactors can provide the specially resolved tracking of products along the multi-function catalyst bed and hence can break down the interaction between the catalysts and the physio-chemical environment.

The first scope of this work is to characterize a newly designed lab-scale profile reactor. The lab-scale profile reactor for n-butane oxidation was designed by Professor Raimund Horn and his team at the Institute of Chemical Reaction Engineering, Hamburg University of Technology. They provided a prototype of the reactor. This profile reactor should find its application in the catalytic investigation of innovative catalyst classes for the selective oxidation of *n*-butane to MAN. So far, only lab-scale profile reactor designs have been published, which do not have to deal with condensable species like MAN. The evaluation will be realized by studying the catalytic behavior of VPP, comparing these results with previously published ones, and simulating the species profiles with an already published kinetic model.

The second scope is the evaluation of new materials synthesized by the group of Professor Glaum via solution combustion synthesis (SCS) and the group of Stephan A. Schunk (hte GmbH) via incipient wetness impregnation (IWI). In cooperation with these two partners, two new catalyst classes have recently been identified as promising candidates by developing two different synthesis concepts. One idea is to condense the most relevant findings of the physio-chemical properties of the VPP catalyst and to develop new V-containing catalyst classes with similar properties. This concept led to the synthesis of unsupported $V_xNb_{1-x}OPO_4$ with an α_{II}-$VOPO_4$/$MoOPO_4$-type structure and nearly XRD-

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amorphous supported $xV(1-x)NbPO_y$ on SiO_2. The second concept involves transforming total combustion platinum group metal (PGM) catalysts into selective oxidation candidates by diluting the metal cations. As a result, a V-free new catalyst class consisting of unsupported crystalline $(Ru_aW_bP_c)O_{2.5+\delta}$ with a ReO_3-related structure and supported catalysts with similar composition by dispersing Ru and W in polyphosphoric acid networks supported on SiO_2 could be realized. This work will conduct a compositional, structural, and performance study to identify the most promising candidates for an in-depth profile reactor investigation.

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2 State-of-the-art

2.1. Profile reactors

In-situ studies are irreplaceable in gaining a deep understanding of a catalytic process, and fix-bed reactors are particularly relevant because of their wide use in industrial applications. A powerful in-situ setup should simultaneously provide various methods for analyzing the state of the catalyst, temperature, and gas-phase composition. Significantly, tracing intermediates along the catalytic bed would help to closely study chemical reaction networks. Traditional tube measurements accomplish this by measuring different catalyst masses at constant gas flow or varying gas velocity. Drawbacks of these measurements are the high time exposure and catalyst material consumption (8 points – 8 reactors) or changed reaction conditions. [1, 15]

Over the last 20 years, the black box behavior of conventional reactor setups has been conquered. William P. Partridge and co-workers developed a novel measurement technique for measuring spatially resolved catalytic behavior via mass spectrometry (SpaciMS). The authors inserted open-end silica capillaries via minimal inversive channels in a monolithic NO_x storage-reduction catalyst, which enabled them to take reaction mixture samples at specific locations along the bed of the catalyst.[19]

The applicability of this approach was demonstrated by a CO oxidation reaction over a Pt/Rh on an alumina catalyst supported on a cordierite monolith. With the help of 3D computational fluid dynamics (CFD) studies, they could show that using thermocouples and capillary probes inside the monolith does not affect reaction behavior.[20]

Lyubovsky et al. [21] studied methane oxidation to syngas by performing profile measurements through a microlith. K-type thermocouples were introduced between microlith screens over the length of the catalytic bed. Gas sample probes were analyzed via online gas chromatography. Due to the limitation of constantly measuring behind microlith screens, a spatial resolution of 2 mm could be reached.

Another approach was chosen by Bosco et al. [22].They designed a profile reactor for the kinetic investigation of the methanation reaction of CO_x and partial oxidation of methane.

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The reactor consisted of a catalytic plate with a window for infrared (IR) thermography and a sampling capillary for gas probing. The capillary was directly situated above the plate and moved with a motor. IR thermography is a powerful tool since the temperature can be measured contactless. Nevertheless, a catalytic plate is far from the concept of fixed beds, which are more industrial-relevant.

Since 2006, the working group led by Raimund Horn and Lanny D. Schmidt [23, 24] has developed the profile reactor concept based on previous work. Initially, they combined the capillary approach with a thermocouple. A very thin thermocouple was inserted in an open-end sampling capillary which enabled them to measure the local temperature at the gas sampling location. The first focus of their work was the investigation of the mechanism for the partial oxidation of methane on Rh-coated α-Al_2O_3 foam monoliths. The chosen reaction has several advantages regarding the characterization for profile reactors. Firstly, the reaction was a gas phase reaction with two reactants (CH_4, O_2) and four possible products (CO, CO_2, H_2, H_2O). This small number of product and reactant species made fast analysis by mass spectrometer possible. Secondly, moving the open-end capillary along the catalytic bed does not affect a monolith catalyst with a stable structure. Compared to Lyuboswsky et al. [21], every position within the catalytic bed could be probed, and a spatial resolution of 0.3 mm could be achieved. Horn et al. [23, 24] compared species and temperature profiles of oxidative coupling of methane with a 2D porous reactor model, including heat and mass transfer. They obtained qualitative and quantitative agreements for syngas stoichiometry for all experimental conditions. Moreover, they performed besides steady-state spatial profiles also spatially resolved transients. The C/O ratio of the reactor feed was changed periodically. Placing the sampling point in the center of the catalytic bed allowed them to extract kinetic parameters from concentration and temperature profiles. This setup revealed how reaction rates depend on temperature, by differentiating between the timescales of chemical interactions (milliseconds) and thermal effects (seconds).

In 2010, Horn et al. [18, 25] changed from an open-end capillary approach to a laser-drilled sampling orifice in a 700 µm fused silica capillary. The new approach allows one to draw the capillary through the catalytic bed and move this capillary reversibly in both directions, making the profile reactor approach feasible for packed catalyst beds. The thermocouple was inserted into the capillary, and the end of the thermocouple was in direct contact with

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the sampling orifice. The reactor setup was designed to reach temperatures of 1300 °C and pressures of 45 bar, which had the claim to be closer to industrial-relevant conditions. Spatial profile measurements have been demonstrated by methane oxidation on platinum and methane oxidative coupling on Li/MgO. Catalysts were supported on α-Al_2O_3 foam. Besides a thermocouple, they alternatively inserted an optical fiber for spatially resolved Raman spectroscopy [18, 25]. However, Raman spectroscopy could only be performed by an open-end capillary at this time, which limited this technique to monolith catalysts. The authors presented their proof of concept by investigating the oxidative coupling of methane over rhodium and platinum-coated foam catalyst.

From 2013 on, they could perform in-situ Raman spectroscopy by studying oxidative dehydration of ethane over MoO_3 catalyst supported on γ-alumina spheres. By using the sampling orifice capillary approach (no open-end) and an optic fiber ending at the location of the sampling orifice, it was possible to take Raman spectra from the radial center of the catalytic bed along the axis. Geske et al. [17] could show the reduction of MoO_3 to MoO_2 as a function of oxygen partial pressure.

Touitou et al. developed a method similar to Raimund Horn and co-workers. With the difference, they drilled even three sampling orifices at a distance of 220 µm. In addition, the orifices were drilled through from both sides. Most probably to avoid a total blocking of probing if spontaneously one sampling orifice is blocked by catalyst particles. [26] Wehinger et al. studied the dry reforming of methane over nickel in a lab-scale profile reactor experimentally and with CFD simulations. They could reproduce with Delgado et al.'s DRM microkinetic model [27] satisfactorily the concentration profiles for CH_4, CO_2, and CO.[28] However, the were not reproducible for H_2 and water. The authors pointed out that these discrepancies can be attributed to possible catalyst dynamics, divergent catalysts, and support composition, as well as to parameters of the microkinetics. Their work showed the importance of profile reactors for CFD studies of fixed bed reactors.

All the improvements mentioned earlier have been implemented in lab-scale reactors. 2018, Dong et al. [29] published a paper with the first pilot-scale fix bed profile reactor. The reactor performed selective oxidation of *n*-butane to maleic anhydride over industrial vanadyl pyrophosphate (VPP). Pilot-scale reactors are essential for getting closer to industrial-relevant conditions. Lab-scale reactors are almost unsuitable for understanding industrial reactors' heat and mass transfer limitations. In combination with kinetic profile

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measurement, particle-resolved CFD and kinetic modeling of intrinsic reaction rates gave a holistic idea about hot-spot behavior and its consequences for maleic anhydride selectivity.

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2.2. Selective oxidation of C1 to C4 alkanes

This section deals with the selective activation of lower gaseous alkanes. The challenges around alkane activation and the requirements of suitable selective oxidation catalysts will be discussed. Followed by the state of the art of selective oxidation for each light alkane towards olefins and oxygenates. Whereby selective oxidation of *n*-butane to maleic anhydride will be reviewed more deeply.

Table 2.1 Industrial and chemical processes at the development stage for the oxidative transformation of light alkanes (Adapted after Ref. [30, 31])

Raw material	Product	Stage of development
Methane	Methanol	Pilot plant
Methane	Ethylene	Pilot plant
Ethane	1,2-Dichlorethane, vinyl chloride	Pilot plant
Ethane	Acetaldehyde	Research
Ethane	Acetic Acid	Research
Ethane	Ethylene	Research
Propane	Acrolein, acetic acid	Research
Propane	Acrylonitrile	Demonstrative plant
Propane	Propylene	Research
n-Butane	Maleic anhydride	Industrial process
Isobutane	Methacrylic acid	Research
Isobutane	Isobutene	Research

Selective oxidation reactions are substantial in the present chemical industry. After polymerization reactions, oxidative transformations use the second-biggest share of chemical production.[32]

The most valuable industrial representatives of selective oxidation processes are propene's selective oxidation to acrolein, propene's ammoxidation to acrylonitrile, ethylene's epoxidation to ethylene oxide, and the oxidation of *n*-butane to MAN. The oxidation of *n*-butane to MAN is the only industrial process using light alkanes as feedstock

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